Correct Method of Ward's Method Analysis Using the hclust Function in R

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1. Introduction

In the March 2021 issue of this journal, we published an article on hierarchical cluster analysis of household expenditure survey data (Kimura and Takabe [1]).

Several readers of the article commented that they were interested in cluster analysis but didn't understand the differences in options for the Ward method in R's hclust function, how to use it correctly, or how to interpret the analysis results.

Osumi [2] points out that, as is common in hierarchical classification, even with the same method name, differences in the selection of similarity/dissimilarity, the stratification procedure, and the use of independently established options can result in completely different output results for the same data depending on the software used.

Murtagh and Legendre [3] also indicate that there are multiple variations of Ward method statistical software available and warn against their use. While the R hclust function subsequently received additional features, including the ward.D and ward.D2 options, the R documentation is not particularly clear for beginners. Inaccurate descriptions can be found in information available on the internet and elsewhere. It is understandable that users may be confused.

This article explains the differences between the ward.D and ward.D2 options of the hclust function currently in use since version 3.0.3, their correct usage, and the basic concepts and characteristics of Ward[4]. Furthermore, it compares the results of applying a program implemented independently by the author based on Ward[4] with the hclust function to an actual data analysis example[1], verifying that the internal processing of the hclust function is consistent with what has been described. The design and implementation of the independently implemented program used here will be explained in detail in the next article. This should serve as a useful reference for concrete examples of implementing mathematical formulas and algorithms in a program. The views expressed in this article are those of the author only and do not represent those of the organization to which the author belongs. Furthermore, any errors or omissions in this text are solely the responsibility of the author.

2. The Correct Use of the hclust Function in R

To state the conclusion first:

① The ward.D option is faithful to Ward[4], and internally, the calculation uses the dissimilarity defined in Ward[4] (hereafter referred to as "ward dissimilarity"). The correct first argument to the function is "half the squared Euclidean distance (= the initial "ward dissimilarity")" (the reason for half the distance will be explained in Section 3 (1)).

② In the case of the ward.D2 option, the internal calculation is based on the square root of "ward dissimilarity". Therefore, the correct first argument is "the value obtained by taking the square root of 'half the squared Euclidean distance' (= the value obtained by taking the square root of the initial "ward dissimilarity")" [5].

If you pass the values to the function as described in ① and ② above, the processing by both options is actually equivalent. In other words, the order and results of cluster aggregation are the same regardless of which option is selected. The only difference is whether "ward dissimilarity" or the square root of "ward dissimilarity" is used as the dissimilarity value handled internally. The resulting Height value indicates the dissimilarity value between aggregated clusters. Therefore, when a dendrogram is drawn from the analysis results of both options, the Height value obtained by taking the square root of value ① will perfectly match the value of ②.

To summarize the conclusion concisely, it is as follows:
When assigning the analysis results to the variable hc by specifying the dissimilarity matrix d and method option in the hclust function,

```
hc <- hclust(d, method="xxxxxx")
```

Here, xxxxxx is assumed to be either ward.D or ward.D2. If the data matrix to be clustered (after preprocessing such as standardization) is DATA, then:

The value to substitute for d is:

When ward.D: (1/2)*dist(DATA)^2

When ward.D2: sqrt(1/2)*dist(DATA)

This is the correct approach according to the ward criterion.

The documentation for the hclust function, Examples EX2, shows an example of using the ward method as follows:

```
hcity.D <- hclust(UScitiesD, "ward.D") # "wrong"

hcity.D2 <- hclust(UScitiesD, "ward.D2")
```

It states that "the ward.D2 example follows the ward criterion, but the ward.D example does not, and the analysis results differ between the two." However, the "wrong" part is the "unsquared, skewed first-order dissimilarity matrix" passed in the ward.D example, not the ward.D option specification itself. If the first-order dissimilarity is correctly set as follows, both examples in EX2 will follow the ward criterion, and the analysis results will be equivalent:

```
hcity.D <- hclust((1/2)*UScitiesD^2, "ward.D")

hcity.D2 <- hclust(sqrt(1/2)*UScitiesD, "ward.D2")
```

(Note that since UScitiesD is a distance matrix between major US cities, the dist function is unnecessary.)

3. Definitions of Terms and Algorithm of the Ward Method

(1) Definitions of "Information Loss" and "Dissimilarity"

In Ward[4], the "loss in information" associated with clustering is expressed by ESS (error sum of squares), and the "increase in information loss" is defined as the "dissimilarity" between clusters. In Ward[4], the paper explains the derivation of the formula and an example of cluster analysis using a one-dimensional example. This paper will explain the meaning of the Ward method in detail in a multidimensional form.

For the i-th cluster i in the aggregation process, let Ni be the number of elements, xij be the j-th element vector of cluster i, and ci be the centroid vector of cluster i. If m is the number of variables in each element, then both xij and ci are m-dimensional vectors.

$$\boldsymbol{x_{ij}} = (x_{ij1}, x_{ij2}, \cdots, x_{ijm}), \ \boldsymbol{c_i} = (c_{i1}, c_{i2}, \cdots, c_{im})$$

If we denote the ESS of cluster i as ESSi, then ESSi is given by the following equation:

$$ESSi = \sum_{j=1}^{Ni} \| \boldsymbol{x_{ij}} - \boldsymbol{c_i} \|^2 \qquad (3\text{-}1)$$

Here, || || is the vector length (Euclidean distance). Next, the "dissimilarity (DSpq)" of cluster p and cluster q is given by the following: Consider the case where clusters p and q are agglomerated to form cluster r:

$$DSpq = ESSr - (ESSp + ESSq) \quad (3\text{-}2)$$

In other words, the "increase" in "information loss" due to the agglomeration of clusters p and q is defined as the "dissimilarity (DSpq)" between clusters p and q.

Furthermore, we transform equation (3-2) using equation (3-1).

$$DSpq = \sum_{j=1}^{Nr} \| \boldsymbol{x}_{rj} - \boldsymbol{c}_r \|^2 - \left(\sum_{j=1}^{Np} \| \boldsymbol{x}_{pj} - \boldsymbol{c}_p \|^2 + \sum_{j=1}^{Nq} \| \boldsymbol{x}_{qj} - \boldsymbol{c}_q \|^2 \right)$$

$$= \{Np \cdot Nq \ / \ (Np + Nq)\} \times \| \boldsymbol{c}_p - \boldsymbol{c}_q \|^2 \quad (3\text{-}3)$$

DSpq can be calculated using the centroid vectors of clusters p and q and the number of elements in each cluster. Initially, each cluster has all elements of 1, so Np = Nq = 1, and therefore Np・Nq / (Np + Nq) = 1/2. This is why the squared Euclidean distance is multiplied by 1/2 in the initial dissimilarity calculation.

(2) Cluster Agglomeration Algorithm

The two clusters with the smallest dissimilarity (DSpq) are agglomerated, and this process is repeated until only one cluster remains. This is the essence of the Ward method proposed by Ward[4].

(3) Update Formula for Efficient Dissimilarity Recalculation

The only dissimilarity that needs to be calculated in each step is the dissimilarity between the new cluster agglomerated in the previous agglomeration step and the other clusters. The dissimilarity between clusters not involved in agglomeration has already been calculated in the previous step. The dissimilarity can be recalculated using formula (3-3), but it is necessary to calculate the centroid vector of the new cluster and the distance from the centroid vectors of the other clusters.

In implementations of tools, in order to minimize the recalculation of dissimilarity, the dissimilarity calculated in the previous step is used to calculate the dissimilarity in the next step. The "Lance-Williams update formula" is used for this purpose. Consider a case where clusters p and q are selected as targets for aggregation in a certain aggregation process and aggregate into cluster r. The dissimilarity (DSrs) between cluster r and the other clusters s can be calculated as follows using DSps, DSqs, and DSpq calculated in the previous process. In the case of Ward's method,

$$\begin{aligned} DSrs = \{1/(Np + Nq + Ns)\} \\ \times \{(Np + Ns) \times DSps + (Nq + Ns) \\ \times DSqs - Ns \times DSpq\} \qquad (3\text{-}4) \end{aligned}$$

Here, Np, Nq, and Ns are the number of elements in clusters p, q, and s, respectively. By using equation (3-4), calculations related to the cluster element vectors become unnecessary [6].

(4) Characteristics of Ward's Method

Ward's method is said to be relatively well-balanced among hierarchical cluster analysis methods and less prone to chain effects. Why is that? I consulted various books on the subject, but to my knowledge, I couldn't find any that clearly stated this. Therefore, this paper will briefly explain the reason.

Equation (3-3) is reproduced below, with some parts omitted.

$$DSpq = \{Np \cdot Nq \ / \ (Np + Nq)\} \times \| \boldsymbol{c_p} - \boldsymbol{c_q} \|^2$$

This equation was the definition of dissimilarity between clusters in Ward's method. The clusters p and q that yield the smallest DSpq are selected as the clusters to be aggregated in the aggregation process. The right-hand side of the equation is a multiplication of two terms. The first term consists only of the number of elements (Np and Nq) contained in the cluster, and the second term is the squared Euclidean distance between the centroids of the two clusters. The second term directly reflects the value of the element vectors contained in the cluster and indicates the magnitude of the difference in features between the clusters. Let's consider the case where there are clusters a, b and clusters c, d, where the second term is the same. In this case, the first term is what determines the magnitude of the dissimilarity.

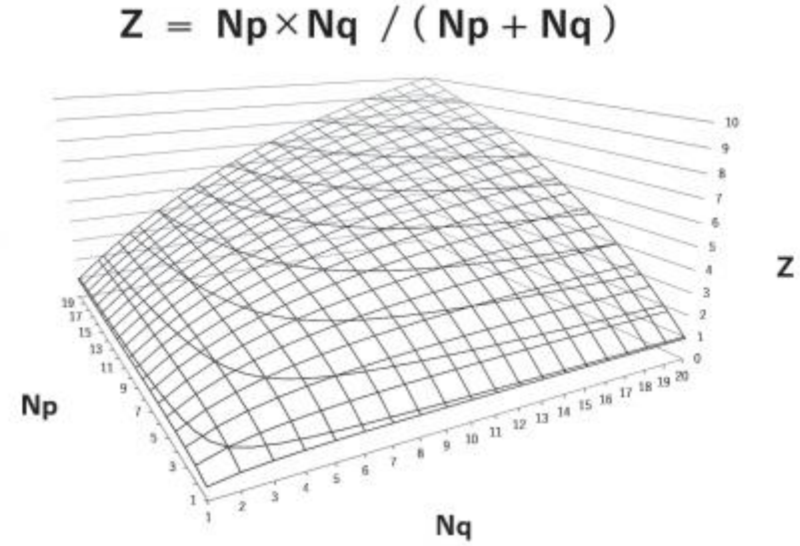

Fig. 1 Graph of the First Term

The first term is a monotonically increasing function with respect to the number of elements. That is, cluster pairs with fewer elements will have smaller dissimilarity, and cohesion will be preferred.

Fig. 1 is a graph of the first term. The effect of the first term creates the characteristic that "cluster pairs with fewer elements are preferentially cohesled."

4. Let's examine the internal processing of the hclust function.

While we've explained the correct usage of the Ward algorithm-related options in the hclust function, many of you may still have lingering doubts, wondering, "This explanation differs from the documentation for the R hclust function. Is it really correct?"

Therefore, to verify the correctness of the conclusions presented in Section 2, we will take a somewhat roundabout approach and compare the internal processing of the author's independently implemented Ward algorithm program with the R hclust function. The independently implemented program is designed entirely in accordance with Ward[4], and uses the initial dissimilarity matrix described in Section 2①. This initial dissimilarity matrix is equivalent to squaring each element of the Euclidean distance matrix created with the R language's dist function and multiplying by 1/2. Starting with this initial Ward dissimilarity matrix, cluster agglomeration proceeds according to the Ward algorithm, and the dissimilarity matrix is updated using equation (3-4).

If the explanation in Section 2 is correct, the results from the custom program and the results from the ward.D option of the hclust function should match. Furthermore, the squared Height value of the analysis result obtained by specifying ward.D2 in the hclust function and using the square root of the "initial ward dissimilarity" as the initial dissimilarity should also match.

5. Verification Experiment: Comparing Analysis Results

The data used for analysis was SSDSE-2020C (Household Expenditure Survey, Food Items) from Kimura and Takabe [1] and the Statistics Center [7].

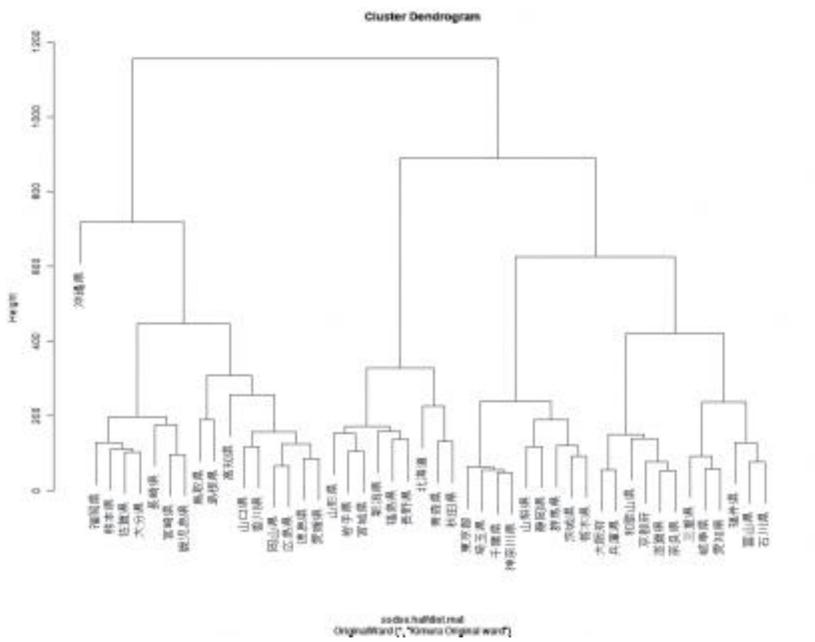

Fig. 2 Results of the Proprietary Program

Fig. 2 shows the results of this proprietary implementation.

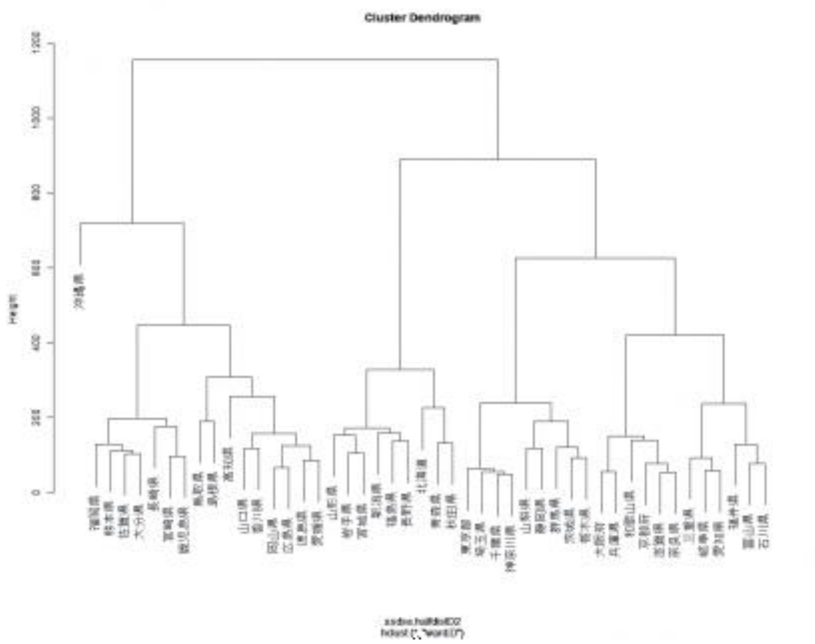

Fig. 3 Results of the hclust function with ward.D specified

Fig. 3 shows the results of using the R hclust function with the ward.D option specified. Figures 2 and 3 are identical, including the Height values on the vertical axis.

Fig. 4 shows the results of using the ward.D2 option with the square root of the "initial ward dissimilarity matrix". The order of cluster aggregation matches that of Figures 2 and 3, but the Height values on the vertical axis are the square roots of the Height values from Figures 2 and 3.

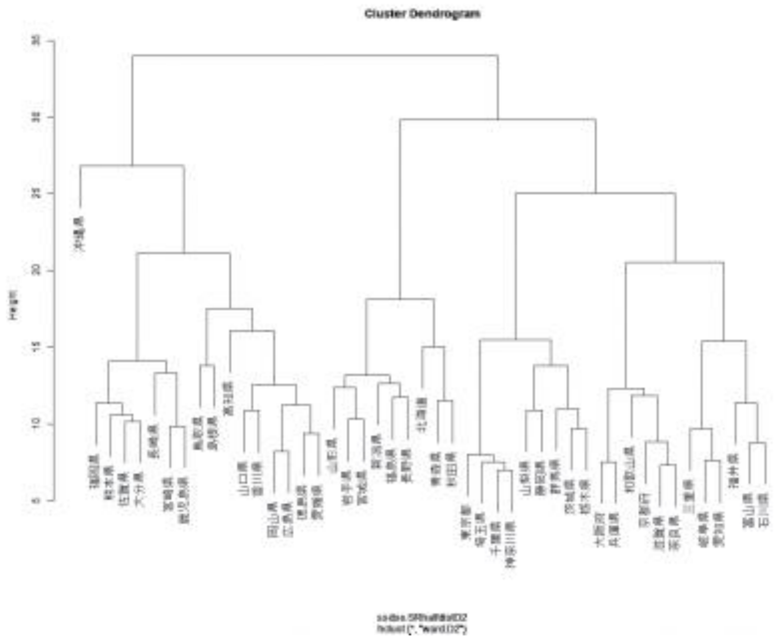

Fig. 4 Results of the hclust function specified in ward.D2

Therefore, Figure 5 shows the dendrogram redrawn using the squared Height values from Figure 4.

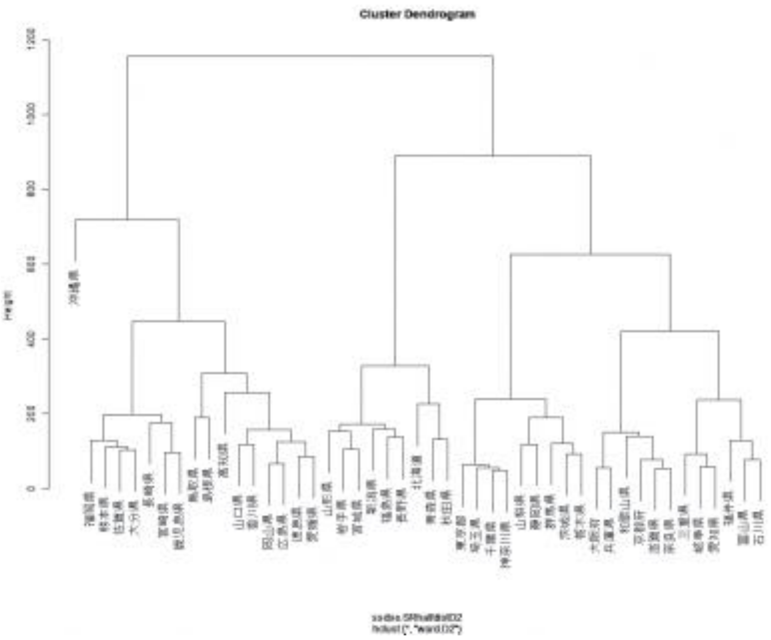

Fig. 5 Analysis results of squaring the Height values from Figure 4

It can be seen that it perfectly matches Figures 2 and 3.

Note that the order of the prefecture names on the horizontal axis in Figure 2 matches the output of the hclust function.

6. The Background to Creating a Custom Implementation Program

Kawabata et al. [5] cite Murtagh and Legendre [3] for their explanation. However, the documentation for the R

hclust function also cites Murtagh and Legendre [3], and their claims appear to differ. This is illustrated in Figure 4: Results of the hclust function with ward.D2 specified.

Figure 5: Analysis results after squaring the Height value in Figure 4.

I was initially quite confused. Let's reconfirm the conclusions of Murtagh and Legendre [3]. The following bullet points summarize my understanding of Murtagh and Legendre's [3] conclusions:

• There are two types of implementation algorithms for the Ward method: implementations based on F. Murtagh's [8] code (hereafter Ward1) and implementations based on Kaufman and Rousseeuw's [9] code (hereafter Ward2).

* Ward1 and Ward2 use different dissimilarity measures internally, and their dissimilarity update formulas are also different. Ward1 uses the squared Euclidean distance for dissimilarity, while Ward2 uses the Euclidean distance.

* However, the Ward1 and Ward2 algorithms are identical, and equivalence can be ensured by appropriately relating their dissimilarity measures.

In other words, Ward1, which uses the squared Euclidean distance as the dissimilarity measure, is the implementation that adheres to Ward's method. At the time of Murtagh and Legendre's publication [3], the R hclust function only had a Ward1 implementation (option method = "ward"). Murtagh and Legendre [3] states that "in the future, the R hclust function will be prepared to include both Ward1 and Ward2 implementations," but it naturally does not describe how the ward.D and ward.D2 options, which were later added to the hclust function, were ultimately implemented. If, as per the R documentation, the correct approach when specifying ward.D is to pass the result of the dist function directly as argument d, then hclust must square d and multiply it by 1/2 internally.

I tried reading the source code of the hclust function to check its internal processing, but the crucial processing was a Fortran subroutine, making direct verification impossible. Therefore, I decided to create my own Ward[4]-compliant program and compare it with hclust.

7. Conclusion

This document provides a detailed explanation of the correct usage of the R hclust function in strict accordance with Ward[4]'s criteria.

When specifying the ward.D option, the dissimilarity of the squared Euclidean distance must be used.

If you pass the output of the dist function without squaring it, you will be analyzing data that has been severely distorted, so caution is necessary.

Note that some commercially available books use the hclust function without multiplying by coefficients such as 1/2 or the square root of 1/2 of the initial dissimilarity. In this case, the resulting Height value will differ from the Ward criterion, which poses a problem when evaluating the Height value itself, but there is no difference in the analysis results if you are only evaluating the cluster aggregation order or aggregation results, so please rest assured.

*References

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